

BYDGOSZCZ, Poland

WMO: 122400

Lat: 53.0964N

Lon: 17.9950E

Elev: 237

StdP: 14.57

Time Zone: 1.00 (EUC)

Period: 02-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) 3.5	(c) 10.1	(d) -0.6	(e) 5.4	(f) 5.2	(g) 4.9	(h) 7.1	(i) 12.0	(j) 27.2	(k) 43.1	(l) 24.6	(m) 40.4	(n) 4.3	(o) 90	(p) 0.561

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	19.6	87.5	67.3	84.0	65.9	80.7	64.7	70.1	81.8	68.7	79.4	67.0	76.6	9.0	150

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
66.5	98.4	73.3	64.7	92.4	72.3	63.0	87.0	71.1	34.3	82.0	33.0	79.1	31.5	76.8	75.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a) 21.5	(b) 18.8	(c) 16.6	(e) -2.3	(f) 92.6	(g) 9.8	(h) 3.5	(i) -9.4	(j) 95.2	(k) -15.1	(l) 97.2	(m) -20.6	(n) 99.2	(o) -27.7	(p) 101.7		
(4) 21.5	(5) 18.8	(6) 16.6	(7) -2.3	(8) 92.6	(9) 9.8	(10) 3.5	(11) -9.4	(12) 95.2	(13) -15.1	(14) 97.2	(15) -20.6	(16) 99.2	(17) -27.7	(18) 101.7		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	48.2	28.8	30.8	37.8	48.1	56.2	62.5	67.2	66.0	57.8	47.8	40.4	33.7
	DBStd	15.16	10.26	8.85	8.36	7.36	7.45	6.23	5.86	5.61	6.07	6.87	6.31	8.53
	HDD50	2648	656	538	383	119	24	1	0	0	7	125	290	505
	HDD65	6403	1121	958	842	508	288	118	45	55	228	533	737	970
	CDD50	1990	0	0	5	63	217	377	532	496	241	56	3	0
	CDD65	272	0	0	0	1	15	44	112	87	13	0	0	0
	CDH74	2891	0	0	0	39	218	554	1073	853	154	0	0	0
	CDH80	833	0	0	0	10	41	155	342	246	39	0	0	0
(14) Wind	WSAvg	7.6	8.6	7.7	8.3	7.8	7.4	7.6	7.1	6.4	6.7	7.4	7.7	8.5
(15) Precipitation	PrecAvg	21.3	1.3	1.0	1.4	1.2	2.1	2.3	3.1	2.5	1.9	1.5	1.4	1.5
	PrecMax	27.2	3.3	2.4	2.8	3.4	3.9	4.4	5.4	6.3	4.7	4.8	4.1	3.1
	PrecMin	15.1	0.2	0.2	0.4	0.1	0.6	0.8	0.3	0.6	0.5	0.2	0.2	0.3
	PrecStd	3.4	0.7	0.5	0.7	0.7	0.8	1.1	1.4	1.3	1.0	1.1	0.7	0.7
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	51.1	52.1	64.2	79.0	84.4	89.8	91.8	91.4	84.6	71.2	57.4	52.2
		MCWB	46.9	46.8	52.2	58.0	64.0	69.6	69.2	69.6	65.4	59.2	54.2	49.2
	2%	DB	46.8	48.3	58.7	71.7	80.2	84.2	87.7	86.0	78.5	66.2	54.0	49.7
		MCWB	44.4	44.5	48.3	56.1	62.7	65.7	67.4	67.6	62.6	57.5	50.6	47.8
	5%	DB	44.2	44.8	53.9	66.5	75.4	80.4	84.0	82.3	73.6	62.2	51.8	46.6
		MCWB	42.2	41.4	46.5	52.7	60.8	63.5	66.5	65.8	61.0	56.0	48.9	44.7
	10%	DB	41.4	41.4	50.2	62.5	71.5	76.6	80.4	78.8	70.0	58.8	49.7	44.3
		MCWB	39.7	39.0	44.6	51.3	58.6	62.1	65.6	64.3	59.6	54.2	47.8	42.6
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	48.0	48.4	53.9	60.6	66.5	71.7	72.1	71.7	67.7	61.4	55.7	50.4
		MCDB	49.8	50.3	61.9	74.1	79.0	84.0	83.5	83.7	82.2	68.2	57.6	51.9
	2%	WB	44.5	44.5	50.4	57.4	64.0	68.6	70.2	69.7	64.8	58.8	51.5	47.5
		MCDB	46.3	47.0	55.5	69.0	75.9	79.6	81.6	81.4	75.2	64.1	53.6	48.9
	5%	WB	42.1	41.6	47.6	54.6	62.0	65.8	69.0	68.0	62.6	56.5	49.5	44.5
		MCDB	43.8	44.0	52.8	64.4	73.0	76.6	79.6	78.0	70.8	61.2	51.1	46.0
	10%	WB	39.7	39.2	45.1	52.2	60.0	63.7	67.4	66.2	60.5	54.5	47.4	42.3
		MCDB	41.2	41.2	49.5	60.7	69.5	73.0	76.7	74.8	68.1	58.5	49.0	43.6
(35) Mean Daily Temperature Range	5% DB	MDBR	8.6	11.3	15.6	20.8	20.5	20.8	19.6	20.3	19.1	14.9	9.3	8.4
		MCDBR	8.5	14.5	22.0	27.7	26.3	26.8	25.4	25.9	25.3	20.8	12.1	9.9
		MCWBR	7.5	11.0	13.9	14.8	12.2	11.8	9.5	10.5	12.2	13.1	9.6	8.5
	5% WB	MCDBR	8.4	12.4	19.2	24.7	23.0	22.5	20.6	21.2	22.1	17.5	11.1	9.5
		MCWBR	7.6	10.0	12.9	14.0	11.8	11.6	9.5	9.9	11.8	11.9	9.7	8.5
(40) Clear-Sky Solar Irradiance	taub		0.311	0.340	0.373	0.418	0.413	0.406	0.437	0.428	0.385	0.365	0.346	0.308
	taud		2.266	2.271	2.242	2.175	2.242	2.290	2.220	2.248	2.346	2.385	2.332	2.292
	Ebn at Noon		207	238	254	257	266	268	256	251	249	228	193	185
	Edh at Noon		20	27	34	42	41	39	42	38	31	24	19	16
(44) All-Sky Solar Radiation	RadAvg		198	417	801	1316	1642	1716	1584	1387	1001	541	220	140
	RadStd		33	64	110	173	175	153	193	129	125	81	30	16

Historical Trends

	DBAvg	Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (35 neighbors)	+0.79	N/A	N/A	N/A	N/A	N/A	N/A	N/A	-279	+128	N/A

Nomenclature: See separate page